

In the United States, as in other industrialized nations, regulatory decisions to protect the environment and public health depend heavily on scientific information. Yet the process of decision-making places unusual strains on science. Knowledge claims are deconstructed during the rule-making process, exposing areas of weakness or uncertainty and threatening the cognitive authority of science. At the same time, the legitimacy of the final regulatory decision depends upon the regulator's ability to reconstruct a plausible scientific rationale for the proposed action. The processes of deconstructing and reconstructing knowledge claims give rise to competition among scientists, public officials and political interest groups, all of whom have a stake in determining how policy-relevant science should be interpreted and by whom. All of these actors use boundary-defining language in order to distinguish between science and policy, and to allocate the right to interpret science in ways that further their own interests. This paper explores the contours of such boundary disputes in the context of controversies over carcinogen regulation. It focuses on the contested definitions and strategic implications of three groups of concepts: trans-science or science policy, risk assessment and risk management, and peer review.

Contested Boundaries in Policy-Relevant Science

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Recent advances in the sociology of scientific knowledge have emphasized the influence of social factors on the content of science. Facts are accepted as authoritative not necessarily because they can be empirically verified, but because they are validated through processes of informal negotiation and can be ranged into frameworks of shared assumptions and inferences. Social processes colour the extent to which pieces of scientific knowledge are perceived as certain, leading to differences in the interpretation of the same facts by scientists of different disciplinary training.¹ In areas of high uncertainty, political interest frequently shapes the presentation of scientific facts and hypotheses to fit different models of 'reality'.² The language in which scientists represent and

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legitimate their claims varies in accordance with the audience to which the representations are made.³ Accordingly, one's impressions of the reliability of scientific knowledge can differ depending on whether one looks at the public language of science or at the private language in which scientists communicate their assessments of certainty to each other.

Though the sociology and philosophy of science both attest to the indeterminacy of knowledge, science has for several centuries maintained its authoritative status as provider of 'truths' about the natural world. To discover some of the reasons, one can turn to further insights from the sociology of science, especially in the work of those who have made the structure of the scientific community their special study. A widely accepted line of explanation emanating from the Mertonian school stresses the shared norms that foster cohesiveness in science, even though its practitioners come from divergent geographic, cultural or linguistic backgrounds.⁴ Other scholars have called attention to the restrictive processes of entry into science, which involve not only an esoteric professional training but screening by numerous 'gatekeepers', such as senior academic colleagues or editors of professional journals.⁵ Cohesion within science is also fostered by 'invisible colleges', 'research circles' or other informal networks that control the diffusion of scientific knowledge.⁶ All of these mechanisms have helped to professionalize science, thereby enhancing the prestige and authority of its practitioners. Finally, the process of peer review, devised by scientists to validate each other's discoveries, reinforces the position of science as an autonomous social institution requiring no external control. Peer review procedures reaffirm not only that scientists maintain strict checks on the quality of each other's work, but that only professional scientists are qualified to pass judgement on the truth or significance of knowledge claims made by other scientists.

Much of the authority of science in the twentieth century rests as well on its success in persuading decision-makers and the public that the Mertonian norms present an accurate picture of the way science 'really works'. Unlike politics, science is 'disinterested' and 'objective' and, unlike religion, it is 'sceptical'. Accordingly, alone among major social institutions, science is believed capable of delivering a true picture of the physical world. Scientists have been quite successful in protecting this claim of exclusivity, jealously guarding their power to define the public image of science, and warding off competing claims by rival disciplines, particularly religion and various manifestations of 'pseudoscience'. Recent studies by Gieryn and others have drawn attention to significant 'boundary' disputes between science and other claimants to

cognitive authority, and have underscored the pivotal role of language in establishing claims about the nature of science.⁷

The authority of science is seriously jeopardized when scientists are called upon to participate in policy-making. Administrative decision-making often requires a probing of the areas of greatest indeterminacy in science. Regulation of risks to health and the environment, in particular, involves issues at the frontiers of current scientific knowledge, where consensus among scientists is most fragile. Both science and regulation seek to establish facts. But the adversarial processes of rule-making employed in the United States presume that 'truth' emerges from an open and ritualized clash of conflicting opinions⁸ rather than from the delicate and informal negotiations that characterize fact-finding in science. US administrative proceedings tend to 'deconstruct' the views held by scientific experts.⁹ This is most clearly evident in formal rule-making, where lawyers use adjudicatory procedures to illuminate areas of weakness in expert testimony, including bias and uncertainty. Even in informal rule-making, however, agencies engage in considerable deconstruction of science in order to persuade reviewing courts that they have considered all possible viewpoints and arrived at a reasoned conclusion. In such settings the exercise of informed professional judgement, central to the performance of science, is not enough to establish an expert's credibility. An administrative process structured according to the rules of law depreciates views that cannot be supported by legally accepted forms of proof, such as empirical observations or published scientific authorities.

Although science is subjected to extreme deconstruction in the US regulatory process, the legitimacy of American regulatory decisions uniquely depends on rational justification, in scientific as well as in economic and legal terms.¹⁰ This means that regulators must eventually present the public with a convincing scientific rationale for actions dealing with technological hazards, marshalling the supporting data and rejecting contrary evidence as persuasively as possible. In the US regulatory system, this public reconstruction of the scientific basis for regulation is one of the key responsibilities of the administrative agencies. Regulations must be backed by the issuing agency's own reading of the scientific evidence, with conflicts and uncertainties resolved in light of the agency's prior experience and its governing legal mandate. Through judicial review the agency head is held accountable for shortcomings in the scientific analysis as well as for the social and political consequences of regulation. Therefore, although scientists are often heavily involved in the early stages of decision-making, the scientific story presented to the public is

finally a creation of the political process, and its ultimate test of validity occurs in the courtroom.

Governmental regulation of risk in the United States thus creates a partial overlap between the processes of scientific and legal inquiry and gives rise to competing claims of authority between science and government, particularly concerning the right to interpret the findings of science. Adherence to the Mertonian norms, coupled with a long tradition of critical peer control, has given scientists an assured basis for claiming cognitive authority. The policy process, however, simultaneously casts doubt on the disinterestedness and the certainty of science. These revealed weaknesses provide grounds for political decision-makers to assert that they have a right to engage in interpreting science, especially in areas that are controversial. A partial removal of cognitive authority to the legal and political arena is seen as the only way of assuring that the interpretation of indeterminate facts reflects the public values embodied in legislation as well as the norms of the scientific community.

With increasing governmental control over science and technology in recent decades, scientists and regulators have both acquired higher stakes in controlling the boundary between policy and science. Policy-makers have an overriding interest in responding flexibly to changing political currents. To the extent that science can be represented as indeterminate, requiring judgement rather than fact-finding, political decision-makers absolve themselves of the need to toe the line on any particular scientific orthodoxy. Emphasizing the indeterminacy in science also gives regulators a better chance of withstanding judicial review, since courts have traditionally been more deferential towards decisions based on discretionary judgements ('policy') than to those based on empirical findings ('science').

For scientists, however, there are serious risks in allowing policy-makers to expose the cognitive indeterminacy of science. The process of deconstruction tends to exaggerate the extent to which science deviates from the Mertonian norms. It suggests, for example, that scientists frequently disagree in their interpretation of data, that experts can be found to support virtually any reading of the evidence, and that the choice among different interpretations is ultimately either arbitrary or else coloured by political interest. Such themes, frequently sounded in regulatory controversies, challenge the view of science as a disinterested search for truth. They are profoundly detrimental to the scientist's self-image, a point that is perhaps under-rated in sociological analyses of reward structures in science. At the same time, exposés of uncertainty and

disunity in science undermine public confidence and raise troublesome questions about whether scientists really deserve the symbolic and material rewards they have claimed from society in this century. Scientists, especially those whose work impinges on policy, thus have much to lose unless they can safeguard the classic normative view of science against charges of excessive indeterminacy.¹¹ To shore up their claims to cognitive authority, scientists have to impose their own boundaries between science and policy, thereby coming into potential conflict with policy-makers pursuing opposing interests.

In this paper I argue that many of the boundary disputes between science and policy are played out in the realm of language. The discourse of risk regulation has provided fertile ground for the creation of new linguistic labels whose primary function is to delimit the boundary between science and the political process. However neutral on their face, these terms are politically charged because they are used to explain or justify the allocation of power and prestige between the institutions of science and government. More generally, boundary-defining language not only serves the immediate interests of social and political groups, but, through the creation of new conceptual categories, opens the way for extending those interests to new or larger domains. The use of such language is an important strategic tool not merely for scientists and political officials, but for all other societal interest groups, such as industry and environmentalists, who have a stake in the way power is distributed among centres of scientific and political authority. A study of the way these disparate interest groups use boundary-defining terms about policy-relevant science reveals their essentially contested character.¹² Subtle shifts in meaning and reference correspond to efforts by each group to ensure that risk disputes are controlled by the institutions most sympathetic to their political interests.

In spite of their competing interests in defining the boundary between policy and science, there are certain rhetorical benchmarks on which all parties agree. It is universally accepted, for example, that 'science' should not be influenced by politics and that judgements as to what constitutes 'good' science should be left to scientists. All agree, as well, that scientists should not be involved in making 'policy'. Scientists themselves are quick to acknowledge that policy concerns should be addressed through the administrative process, *after* science has provided all the relevant evidence to the agencies. But while no one doubts that science should be done by scientists and policy by policy-makers, the problem for each interest group is to draw the dividing line between science and policy in ways that enlarge its own control over social

decisions.¹³ Competition among these groups leads to differing definitions of the point at which the autonomy of science ends and the role of political decision-making begins.

This paper examines three contested boundaries that have acquired special visibility in the regulatory discourse of the 1980s. The first concerns the attempt to define 'science' itself by distinguishing it from related concepts such as 'trans-science' or 'science policy'. The second boundary is the uncertain dividing line between 'risk assessment' and 'risk management' around which much of the American regulatory debate has been organized in recent years. The third is the notion of 'peer review', a term used only lately in relation to science in the policy environment, but already endowed with different readings by different political interests. While the first two disputes cover roughly the same terrain – how to distinguish science from policy – the debate about peer review seems to break new ground, namely, the selection of procedures for deciding scientific controversies relevant to policy. I suggest, however, that the new rhetorical emphasis on peer review, as well as the varying meanings assigned to the term, reflect the same political conflicts that underlie the other two boundary disputes. Talking about peer review instead of science policy or risk assessment thus represents a kind of rhetorical frame-shifting that is not untypical in political discourse.

Beyond Trans-Science

Regulatory problems demanding cooperation between scientists and public officials have been part of the political agenda in the industrialized countries for well over a century. The industrial revolution brought in its wake a variety of health, safety and environmental problems that required legislative or judicial attention, and experts played a growing role in shaping governmental responses. Until the 1960s the cooperation between experts and public decision-makers appeared fairly unproblematic. Scientists provided government with technical information and advice, while administrators and judges formulated legal or policy decisions. Few asked whether the scientists were performing purely scientific assessments or whether their deliberations were entirely free from political influence.

For science in the policy context, the age of innocence ended in the early 1970s. The environmental movement gathered momentum, focusing attention on hitherto unsuspected risks to health and the environment.

The United States, along with most other industrial nations, enacted a spate of new legislation in order to prevent cancer, birth defects, degradation of the environment, loss of wildlife and depletion of natural resources. These preventive policies placed unprecedented demands on the capacity of science to predict future harm. Fed by images of impending environmental disaster, the public turned to science for more sophisticated methods of identifying and measuring risk. Science responded with a new emphasis on toxicological testing and increased use of predictive mathematical models. But this shift of scientific attention to the unknown, and possibly unknowable, effects of technology highlighted the intuitive, subjective and uncertain underpinnings of much of the advice that scientists provide to government. Moreover, the increasingly adjudicatory style of decision-making in the United States forced scientists to articulate their reservations about their technical assessments and generated questions about the coherence or reliability of policy-relevant science.

An important turning point for the relationship between scientists and regulators came in 1972 with the publication of Alvin Weinberg's article on 'Science and Trans-Science'.¹⁴ In this article, Weinberg alerted both the scientific and the regulatory community to the existence of a grey zone between science and policy, characterized by questions 'which can be asked of science and yet which cannot be answered by science'.¹⁵ Weinberg called this grey area 'trans-science', a term which, like C. P. Snow's 'two cultures', struck an unexpectedly responsive chord among a wide audience. It seemed to capture a distinction that many had been groping for in trying to understand policy controversies based on science. But subsequent elaborations of Weinberg's analysis suggest that the versatility of the term 'trans-science' helped assure its popularity. 'Trans-science' became a skeleton key by which different interest groups could attempt to unlock different doors in keeping with their specialized political objectives.

The concept of 'trans-science' served a strategically crucial purpose for scientists interested in reaffirming their professional authority. It suggested that the cognitive indeterminacy uncovered by the regulatory process was not intrinsic to science, but rather lay beyond or outside science. Conversely, an important corollary of Weinberg's analysis was that science proper is untouched by the uncertainties and expert conflicts that necessarily occur in trans-science. Thus, while regulators could be allowed considerable leeway to probe, and even decide, trans-scientific issues, science itself should remain the undisputed preserve of scientists, to be performed, evaluated and tested in accordance with the internal norms and procedures of the profession. Weinberg himself was careful

to draw out some of the policy implications of his boundary-defining terminology, particularly in relation to the structure of the decision-making process. The following passage is especially revealing:

If the question is unambiguously scientific, then the procedures of science rather than the procedures of law are required for arriving at the truth. Where the questions cannot be answered from existing scientific knowledge or from research which could be carried out reasonably rapidly and without disproportionate expense, then the answers must be trans-scientific and the adversary procedure seems therefore to be the best alternative.¹⁶

In other words, the deconstructionist techniques of establishing truth might be perfectly appropriate for trans-science, but they are out of place within the halls of genuine science.

A recent revisitation of the same boundary problem by Weinberg reveals even more clearly his desire to shield science against the taints of subjectivity, bias and disharmony that it acquires in the policy environment:

No one would dispute that judgments of scientific truth are much affected by the scientist's value system when the issues are at or close to the boundary between science and trans-science. On the other hand, as the matter under dispute approaches the domain of science, most would claim that the scientist's extrascientific values intrude less and less.¹⁷

The passage illustrates an unswerving commitment to the Mertonian ideal of science, a position Weinberg underscores by blisteringly attacking the sociologists of knowledge who, in his view, have presumed to project a contrary image:

At least the more extreme of the sociologists of knowledge claim that using traditional ways of establishing scientific truth – by appealing to nature in a disciplined manner – is *not how science really works*. Scientists are seen as competitors for prestige, pay, and power, and it is the interplay among these conflicting aspirations, *not the working of some underlying scientific ethic*, that defines scientific truth. To be sure, these attitudes toward science are not widely held by practicing scientists; however, they . . . nevertheless exert important influence on other institutions . . . that ultimately influence public attitudes toward science and its technologies.¹⁸

Weinberg's concern quite clearly is that the uncertainties uncovered by the regulatory process will provide ammunition for those wishing to impugn the existence of an 'underlying scientific ethic' or to present a misleading picture of 'how science really works'. To guard against such attacks on 'the core of science',¹⁹ Weinberg is prepared to concede a great deal of ground at the boundary between science and trans-science. His article

on trans-science had already concluded that this boundary is 'elusive'. In his more recent work, as noted below, Weinberg appears willing to admit that much of the so-called 'science' of probabilistic risk analysis – an increasingly important source of conflict in regulatory proceedings – should be relegated to the trans-scientific side of the boundary. Indeed, Weinberg goes so far as to suggest that one should 'define a new branch of science, called regulatory science, in which the norms of scientific proof are less demanding than are the norms in ordinary science'.²⁰

For non-scientific actors in the regulatory process, what is at stake in the boundary dispute between science and policy is not so much the image of science or the scientist as the question of institutional power. In a given regulatory proceeding, who should determine how the boundary is drawn between science and trans-science? And once this line is established, who should decide the controversial trans-scientific issues and by what procedures? Weinberg's original article seemingly viewed the boundary-defining function as a prerogative of scientists. In discussing the scientist's place in the 'republic of trans-science', Weinberg commented that 'though the scientist cannot provide definite answers to trans-scientific questions . . . , he does have one crucially important role: to make clear where science ends and trans-science begins'.²¹

A somewhat different position has been advanced by administrative lawyers, who have viewed Weinberg's analysis as a rationale for expanding the role of law and legal processes in decision-making at the frontiers of scientific knowledge. Within the growing body of legal scholarship on science and the regulatory process, Thomas McGarity's work on discretionary decision-making in the Environmental Protection Agency (EPA) and the Occupational Safety and Health Administration (OSHA) has proved particularly influential.²² McGarity was drawn to an exploration of the boundary between science and policy through a study of efforts by the federal agencies to control chemical carcinogens in the 1970s. This regulatory endeavour presented the agencies with seemingly intransigent technical problems. Federal regulators recognized that the existence of a cancer risk to humans can seldom be established by means of direct evidence, since there is a paucity of reliable epidemiological data linking specific chemicals to cancer. As a result, regulatory determinations that a chemical may pose a cancer risk depend mainly on secondary evidence, such as animal tests and studies of a compound's capacity to induce mutations. But the interpretation of such data is fraught with uncertainty and expert disagreements, and the regulatory outcome seems to depend less on science than on the institutions and procedures that are used to resolve the proliferating technical conflicts.

Over the past fifteen years a number of federal agencies have taken up the challenge of creating a principled framework for regulating carcinogens. EPA and OSHA took the lead on this issue in the 1970s because of their pre-eminent role in controlling toxic substances. Later their efforts were paralleled or supplemented by the initiatives of the Interagency Regulatory Liaison Group (IRLG), which was active during the Carter administration, and by the Office of Science and Technology Policy (OSTP).²³ The approaches taken by each of these agencies have differed in form and scope. EPA and IRLG issued their principles of carcinogenic risk assessment in the form of relatively flexible guidelines, while OSHA attempted to cast its generic cancer policy in the shape of more formal regulations. OSTP tried to steer clear of making policy judgements in its risk assessment principles, whereas the other agencies acknowledged that policy determinations were needed in order to bridge areas of uncertainty. With regard to science, however, all of these frameworks adopted a basically deconstructionist approach. As a result, the number and complexity of principles used to define the process of risk assessment have increased over time. The agencies have become more and more sensitive as well to the need to characterize the uncertainties in carcinogenic risk assessment, and have developed elaborate methodological caveats for dealing with uncertain issues.

McGarity's analysis of carcinogen regulation by OSHA and EPA focused on a class of issues which he described as 'science policy' because 'both science and policy considerations play a role in their resolution'.²⁴ It is useful to look at his detailed elaboration of the science policy concept, since it too has gained currency in the discourse of health and safety regulation. Under the heading of science policy, McGarity included three broad types of issues: first, questions that are cast in scientific terms, but are inherently unanswerable by science for practical or moral reasons;²⁵ second, questions that cannot be answered because of insufficient scientific data, but are theoretically subject to resolution given adequate time and resources; and third, questions characterized by expert disagreements about either the interpretation of scientific studies or the inferences drawn from them. A careful reading of Weinberg's work suggests that he was well aware of such differences in the typology of 'unanswerable' questions and meant to subsume them all within the phrase trans-science.²⁶ Thus, it is difficult to argue that in purely conceptual terms McGarity's phrase 'science policy' marks a clear advance over the notion of trans-science. Nevertheless, the newer label continues to be widely used in discussions of science-based regulatory decisions. The explanation seems to lie largely in the term's connotations, which carry

different rhetorical and strategic implications than those associated with the term 'trans-science'.

Perhaps the most significant feature of the phrase 'science policy' is that it has prescriptive overtones that are favoured by analysts trained in administrative law. Science policy, after all, can be regarded as just another specialized subfield of policy which can be governed by the normal institutional and procedural controls of American administrative law. This point of view dominates McGarity's own writing. For example, he recognizes as clearly as Weinberg that the boundary between science and science policy is likely to be disputed, so that 'the regulator will need a mechanism for deciding whether an issue is in fact a science policy issue'.²⁷ The mechanism McGarity proposes, however, is 'direct oral testimony with cross-examination', a legalistic solution, which leaves ultimate authority for distinguishing between science and science policy in the hands of a judge or administrator rather than under control of scientific experts.

In his essay on EPA and OSHA, McGarity examined the procedures that might be appropriate for resolving science policy questions. In keeping with the author's disciplinary biases, the article unquestioningly assumed that these procedures should be derived from the existing framework of administrative law, a framework that favours a principled, judicially reviewable mode of decision-making. The focus of McGarity's analysis was a comparison of two different approaches to deconstructing the scientific basis of policy decisions: the innovative 'generic' procedures developed by EPA and OSHA in the late 1970s and the classic adjudicatory model of establishing the truth in disputes over scientific facts. The former approach sought to resolve certain issues relating to chemical carcinogenicity at a generic level, while the latter aimed at establishing the effects of particular substances through individualized, trial-type proceedings. The important point here is that McGarity considered *all* science policy issues as suitable for resolution through one or the other approach, each calling for a public deconstruction of scientific evidence followed by reconstruction according to established judicial or administrative rules. The open question for McGarity was merely how to choose between the two processes. Should the procedures adopted by the agencies be generic or particularized, informal or adjudicatory?

The strategic importance of these assumptions became clearly apparent in a controversy over formaldehyde, which EPA attempted to regulate during the ill-fated administration of Anne Gorsuch.²⁸ Formaldehyde met EPA's and OSHA's principal generic criteria for treating chemical substances as potential human carcinogens. An inhalation study in rats

showed that formaldehyde is an undisputed animal carcinogen, producing nasal tumours in a high percentage of the exposed rodents at high doses. The compound also responded positively in a variety of tests for detecting mutagenicity. Nevertheless, John Todhunter, then EPA's assistant administrator for toxic substances, determined that formaldehyde did not present a significant cancer risk to humans. He reached this conclusion by assessing the properties of formaldehyde that seemed to weigh against a finding of carcinogenicity: negative bioassays in mice and other animals; negative results in studies using routes of administration other than inhalation; the absence of positive epidemiological findings; and the apparent reversibility of cell damage caused by exposure to formaldehyde at low doses or for short durations. These pieces of evidence led Todhunter to argue that the effects of formaldehyde on rats should be regarded as species-specific, hence not inherently problematic from the standpoint of human health.

In building his case on formaldehyde, Todhunter exercised the US administrator's privilege of picking and choosing from the scientific record those bits of data that best supported his policy decision. The problem, however, was that a backdrop of prior EPA rule-making constrained Todhunter's freedom to exonerate formaldehyde from the suspicion of carcinogenicity. In particular, EPA's pre-existing guidelines for assessing carcinogenic risk indicated that regulators should attach much greater weight to positive than to negative studies when evaluating conflicting data on carcinogenicity. Todhunter's deviations from this cardinal principle drew outraged reactions from environmentalists, Congress and members of the scientific community.²⁹ One vocal and highly effective critic, Nicholas Ashford, took the administrator to task for his mishandling of science policy.³⁰ By invoking this concept Ashford defended a particular approach to decision-making which, in the case of formaldehyde, might well have led to a different regulatory outcome from that intended by Todhunter.

Ashford's definition of 'science policy' highlighted the policy component of decisions at the borderline of science and policy:

The term 'science policy' denotes issues that are grounded in scientific analysis but for which technical data are insufficient to support an unequivocal scientific conclusion. The ultimate resolution of these issues depends on determinations of social policy.³¹

This definition suggests that disagreements within science – signalled by the absence of unequivocal conclusions – suffice to characterize disputed technical issues as science policy. Having emphasized the

non-technical dimensions of science policy, Ashford also stressed the need to maintain traditional administrative values in making science policy decisions, for example, openness and adequate public notice, especially when departing from prior agency propositions.³² Such procedural requirements, Ashford charged, were violated by Todhunter's private and unilateral decision to downplay the cancer risk presented by formaldehyde.

This argument sounds compelling if one accepts the major premises underlying Ashford's and McGarity's analyses of science policy: that issues characterized by insufficient data or expert disagreements should not be classified as pure science but as science policy; that procedures for dealing with such issues should be defined by agencies rather than scientists; and that regulators should control decisions about whether particular issues should be categorized as science or science policy. It follows reasonably from these assumptions that changes in science policy should be made in accordance with the legal requirements that apply to administrative policy-making in general.

Ashford went even further. He suggested that policy considerations can override science altogether in resolving disputes about science policy:

In appropriate circumstances, of course, the agency may depart from scientific opinion on science policy issues. These issues do, after all, involve policy determinations, and accordingly should be made by the governmental entity charged with reflecting the will of the people through the execution of a congressional mandate.³³

Such assumptions, however, have by no means gained universal acceptance. In particular, the chemical industry, which stood to lose most heavily from a strict application of the generic policies developed by EPA and OSHA, never agreed that the key technical issues in regulating carcinogens should be characterized as 'science policy'. Instead, industry spokesmen have always argued that carcinogen regulation is based on 'science', and that agencies should be careful not to subvert this science through technically naive or politically motivated administrative rules. Consistent with this position, the American Industrial Health Council (AIHC), a trade association formed to consolidate the chemical industry's opposition to generic regulation of carcinogens, vociferously attacked the government's proposed policies for identifying and classifying carcinogens, including OSHA's cancer policy and EPA's proposed generic approach to regulating airborne carcinogens. AIHC charged that science and policy are impermissibly mingled in all attempts to regulate carcinogens generically. OSHA's cancer policy attracted especially

adverse comment. AIHC denounced it as an attempt to 'freeze science' through a 'cookbook approach' to reading scientific evidence.³⁴ As industry saw it, even controversial issues, such as the interpretation of animal data or techniques for extrapolating risk from high to low doses, were essentially scientific in character and should have been addressed by scientists rather than administrative agencies.

The formaldehyde controversy sheds interesting light on the scientific community's response to such partisan debates about the proper way to resolve disputes at the boundary of science and policy. One might expect *a priori* that scientists would hesitate to accept the full implications of Ashford's position, particularly his suggestion that social policy considerations should, if necessary, take precedence over scientific opinion in making regulatory choices. Given the institutional interests of science, one might expect scientists to find AIHC's rhetoric rather more appealing, since it caters to the image of science as an élite and apolitical source of cognitive authority. Yet in the debate over the carcinogenicity of formaldehyde a number of US scientists with impeccable professional credentials sided with Ashford and the environmentalists rather than with Todhunter and the chemical industry. Their use of language indicates how they reconciled this position with appeals to the authority of science.

Testifying before Congress in 1983, Bernard Weinstein of Columbia University expressed his conviction that the regulation of potential carcinogens 'must be firmly rooted in scientific principles'.³⁵ As examples of such principles, however, he cited many of the generic propositions adopted by EPA, OSHA and IRLG for assessing carcinogenic risk. Weinstein thus underplayed the policy content of the cancer principles and focused instead on the claim that they were all grounded in sound science. He stressed that there was no scientific support for deviating from any of these principles, as Todhunter apparently had done, and concluded that 'our understanding of the mechanism of action of environmental carcinogens... is in a sufficient state of flux that it would be premature to alter the existing, well-established guidelines for detecting potential carcinogens'.³⁶

Unlike Weinstein, Norton Nelson of New York University, who testified at the same set of hearings, referred to the principles as the federal government's 'cancer policy'. But Nelson left little doubt that he regarded this 'policy' essentially as an expression of scientific consensus. He noted that the principles underlying cancer testing and evaluation at EPA had found wide acceptance not only in the United States but among other national and international institutions concerned with regulating carcinogens.³⁷ From this vantage point, Nelson criticized Todhunter's

analysis of formaldehyde as a striking departure from the prevailing scientific consensus, calling it 'the sort of a document that one would not expect an objective scientist to produce'.³⁸ Nelson admitted that advances in understanding the biology of cancer causation had created the need for a thorough revision of federal cancer policies. But he emphasized that such review and updating should be undertaken collegially by an accredited scientific body, preferably the National Academy of Sciences, so as to produce a more up-to-date consensus. Thus, both Nelson and Weinstein implicitly redefined the problematic issues in carcinogen regulation as fundamentally scientific. They were then able to challenge Todhunter's analysis of formaldehyde seemingly on the basis of science alone (and on their personal authority as scientists). Neither scientist endorsed the unpalatable notion that the dispute was a matter of science policy on which science had little definitive guidance to offer.

Risk Assessment and Risk Management

Some federal agencies, including EPA, were regularly performing risk assessments for chemical carcinogens by the late 1970s. Over the next few years, support for the concept of risk assessment widened, helped along by pressure from the chemical industry and the anti-regulatory temper of the Reagan administration. Critiques of the generic approach to carcinogen regulation provided much of the intellectual justification for the spread of risk assessment methodologies. Industry alleged that generic procedures slighted scientific evidence in the interests of a heavy-handed and overly conservative policy of risk management. According to this view, generic policies were fundamentally flawed because they overlooked some of the key factors bearing on the degree of risk, most notably the relationship between dose and response and the extent of public exposure to particular toxic substances. As a result, generic policies were incapable of distinguishing between chemicals posing relatively high and comparatively trivial risks to human health. Industry argued that these systematic errors could be corrected only through comprehensive assessment of the quantitative as well as the qualitative dimension of chemical risks.

Supporters of risk assessment won a major legal victory in a lawsuit challenging the new occupational safety and health standard proposed by OSHA for benzene, a substance known to cause leukaemia in humans exposed to high doses. Acting in accordance with its generic cancer policy,³⁹ OSHA classified benzene as a high-risk substance and

proposed to reduce workplace exposure to the lowest feasible level. Also consistently with the cancer policy, OSHA declined to perform a quantitative risk assessment for benzene. The petroleum industry objected to the proposed standard on several grounds and took the agency to court. The case was ultimately decided by the Supreme Court, where a plurality of the Justices held, in effect, that OSHA's failure to carry out a risk assessment rendered the proposed standard invalid.⁴⁰ According to the plurality opinion, OSHA's governing statute required the agency to demonstrate that there was a significant risk to workers at or below the existing exposure limit prior to issuing a new standard. The agency had conspicuously failed to make such a showing.

In industry's political agenda, however, the demand for formal risk assessment was part of a larger strategy to remove risk assessment from the control of agency scientists and bureaucrats, whom industry regarded on the whole as captive to pro-regulatory interests. AIHC, in particular, issued a series of position papers calling for a new institution to take over the task of doing risk assessments for the regulatory agencies.⁴¹ Specifically, AIHC argued that the scientific integrity of risk assessment could be preserved only by delegating the task to a panel of independent scientists, preferably under the auspices of the National Academy of Sciences (NAS). The thrust of the proposal was to remove all marginally scientific issues involved in carcinogen regulation to the jurisdiction of an expert panel that would have little day-to-day contact with the legal and political dimensions of the regulatory process. But the proposal's political force derived from its appeal to the institutional authority of science, an appeal that many found reassuring after the endless scientific uncertainties laid bare by regulatory deconstruction.

Congress responded to AIHC lobbyists by asking NAS to study mechanisms for improving the quality and credibility of the science used in risk decisions. The National Research Council's Commission on Life Sciences formed a committee to examine risk management practices in the federal government, with the following main objectives in view:

To assess the merits of separating the analytic functions of developing risk assessments from the regulatory functions of making policy decisions.

To consider the feasibility of designating a single organization to do risk assessments for all regulatory agencies.

To consider the feasibility of developing uniform risk assessment guidelines for use by all regulatory agencies.⁴²

The focus on separating science from policy and on institutional reorganization both catered to the chemical industry's interest in removing much of the responsibility for interpreting technical information from the control of individual agencies.

One might well wonder at this point what had happened to the insights about policy-relevant science captured by Weinberg or McGarity. Political demand for separating science and policy seemed to deny the existence of any ambiguous border area where the two kinds of issues are inextricably joined. In fact, awareness of the fuzzy boundary between science and policy soon surfaced within the NAS committee. The problem was clearly laid out in a working paper by Lawrence McCray, project director for the risk management study.⁴³ McCray pointed out that experts familiar with carcinogenic risk assessment saw all but a handful of the analytical steps as involving a mixture of science and policy. Users of the technique agreed that very little in a typical risk assessment could be labelled as pure science. McCray's findings decisively influenced the committee, which ultimately refused to endorse a single, central board for regulatory risk assessment. Conceding that policy choices are involved in risk assessment,⁴⁴ the committee concluded that it was unrealistic to separate this process institutionally from the process of deciding how to manage risks.

Yet the NAS report also made numerous rhetorical bows to the notion that science can be separated from policy through sufficiently rigorous analysis. The idea was implicit in the committee's decision to characterize regulation in terms of two distinguishable processes: risk assessment ('the characterization of the potential adverse health effects of human exposures to environmental hazards') and risk management ('the process of evaluating alternative regulatory actions and selecting among them').⁴⁵ The terminology alone suggests that the scientific and technical aspects of decision-making can be isolated from the socio-political ones. The report played further on this theme by exhorting the agencies to keep the two processes conceptually distinct:

We recommend that regulatory agencies take steps to establish and maintain a clear conceptual distinction between assessment of risks and consideration of risk management alternatives; that is, the scientific findings and policy judgments embodied in risk assessments should be explicitly distinguished from the political, economic, and technical considerations that influence the design and choice of regulatory strategies.⁴⁶

In a similar vein, the report noted:

Even the *perception* that risk management considerations are influencing the conduct of risk assessment in an important way will cause the assessment and regulatory decisions based on them to lack credibility.⁴⁷

Before an agency decides whether a substance should or should not be regulated as a health hazard, a detailed and comprehensive written risk assessment should be prepared and made publicly accessible. This written assessment should clearly distinguish between the scientific basis and the policy basis for the agency's conclusions.⁴⁸

A frequent deficiency of agency risk assessments is the failure to distinguish between scientific and policy considerations in risk assessment.⁴⁹

These remarks and recommendations all point to a powerful reluctance on the part of at least some NAS committee members to admit that science and policy might, at the boundary, be very difficult to distinguish from one another.

In all, the NAS report left the agencies in something of an intellectual quandary. They were instructed to distinguish as far as possible between the assessment and the management of risk. At the same time, they were informed that the two stages are inextricably linked, making any formal institutional separation of the two functions impractical. In view of its indecisiveness, it is not surprising that the report failed to impose consistency on the use of the key terms 'risk assessment' and 'risk management' by the policy community. In spite of the NAS committee's assertions to the contrary, agency officials saw some advantages in treating risk assessment as a wholly scientific enterprise, at least for rhetorical purposes. Emphasizing the objectivity of the procedure offered a means of persuading the public that regulatory decisions are based on a core of rational analysis and of enhancing public confidence in the impartiality of agency decisions.

This analysis may help explain an early insistence on the objectivity of risk assessment by William Ruckelshaus, twice the administrator of EPA, and the person credited with restoring the agency's credibility following Gorsuch's tenure in that office. Ruckelshaus conceded that in an imperfect world risk assessment may be influenced by extraneous factors such as the pressure of the regulatory timetable and limitations on the agency's resources. Nevertheless, he argued in a widely publicized address to the National Academy of Sciences that:

Despite these often compelling pressures, risk assessment at EPA must be based only on scientific evidence and scientific consensus. Nothing will erode public confidence faster than the suspicion that policy considerations have been allowed to influence the assessment of risk.⁵⁰

These remarks, of course, ran directly counter to the NAS committee's judgement that the analytical choices an agency makes in risk assessment must be influenced in part by policy considerations, especially the all-important decision about how conservative the agency wishes to be in determining a risk to public health.

As an experienced administrator, Ruckelshaus could not long ignore the fact that his speech to the National Academy was based on an unrealistic view of the objectivity of risk analysis. Indeed, EPA's own staff scientists maintained a much more sceptical view of the process, characterizing it as no more certain than a circumstantial murder trial, and possibly no better than pulling numbers out of thin air.⁵¹ Ruckelshaus eventually withdrew from his endorsement of a formal separation between risk assessment and risk management, conceding that values influence the former as well as the latter. Characteristically, the alternative he put forward derived from the traditions of the American administrative process and accorded a central role to the deconstruction of science:

I think we need to dig up. We have to expose the assumptions that go into risk assessments. We have to admit our uncertainties and confront the public with the complex nature of decisions about risk.⁵²

Following this analysis, Ruckelshaus and others have suggested that the agencies should be less concerned with the process of risk assessment and more with the transmission of their analytical efforts to the public. These views have led to greater emphasis on 'risk communication' – in other words, on the processes by which administrative agencies publicly reconstruct their scientific rationale for regulating risk.

In spite of Ruckelshaus's public turnaround, the notion that risk assessment can be compartmentalized as an objective, value-free exercise continues to hold powerful appeal. Yet the contrasting view that risk assessment is subjective and highly discretionary also has many adherents. As a result, inconsistent descriptions of the process are sprinkled throughout the policy literature. A recent report written for the Secretary of Health and Human Services (HHS) on assessing and managing toxic substances illustrates the phenomenon.⁵³ The report begins by stressing the fluid and hybrid character of risk assessment:

The process of risk assessment, a process that is still very much in transition or evolution, is often a mixture of scientific facts, consensus, assumptions, and science policy decisions (i.e., policy statements made by agencies to resolve points of current controversy).⁵⁴

But a later passage provides a curiously contradictory description:

The actual process of assessing the risk should remain independent of the value judgments that are necessarily associated with the broader issues of managing the risk. Insuring the quality and independence of the scientific component of risk-management decisions is vital to the public's ability to weigh properly the social and philosophical values intrinsic in such decisions.⁵⁵

Here the thrust is again towards analytically separating risk assessment from risk management and keeping science distinct from social and philosophical values in decision-making about risk.

It appears, then, that recasting the discourse of health and safety regulation in terms of risk assessment and risk management has not resolved the basic dilemmas that arise in the process of using science for policy. True, the phrase 'risk assessment' connotes a form of inquiry that is more like science than policy. Yet everyone concedes that there are elements of science policy or trans-science in risk assessment, and, if McCray's analysis is valid, such elements may even predominate over those that are purely scientific. If risk assessment involves a mix of scientific and policy considerations, then the use of the term fails to resolve the institutional questions posed by Weinberg and others. Who should carry out such analyses and under what formal constraints? What, in particular, is the role of science in risk assessment, and how should responsibility for providing authoritative assessments of risk be allocated among the scientists, policy-makers and political interest groups?

The NAS report gave the regulatory agencies some pragmatic guidance in answering these questions. As mentioned earlier, it put to rest the idea that all risk assessments should be performed by a central scientific panel independent of the regulatory agencies. This means that the agencies retain primary responsibility for developing risk assessment methodologies and applying them to specific cases. The NAS report also recommended that guidelines for drawing inferences from risk data be developed uniformly for all federal agencies by a single expert board,⁵⁶ but there appears little likelihood that an independent board along the lines recommended by the National Academy will be constituted in the foreseeable future.

With significant institutional reform ruled out, the development of risk assessment guidelines has been taken over by agencies having a particular interest in this area. For example, in March 1985 the Office of Science and Technology Policy issued a lengthy document to serve as 'a framework for regulatory agencies in assessing cancer risks from chemicals',⁵⁷ and, in November 1984, EPA issued its own proposed

guidelines for assessing carcinogenic risk.⁵⁸ The OSTP document sought to reflect the current scientific consensus on carcinogenic risk assessment, and therefore explicitly disavowed all attempts to formulate policy. By contrast, EPA needed risk assessment guidelines for use in its various regulatory programmes and thus adopted positions with clear policy implications. In its final guidelines the agency included phrases (for example, 'the agency takes the position that') to indicate more clearly where it was making a policy decision.⁵⁹

The EPA and OSTP guidelines represent more advanced thinking on the subject of chemical carcinogenesis than did the generic policies of the 1970s. But it would be difficult to argue that these initiatives fundamentally altered the balance of power between scientists and policy-makers with respect to carcinogen regulation. Both sets of guidelines called for a principled analysis of the uncertainties in the data. Disclosing these uncertainties, however, tends to place considerable discretion in the hands of the ultimate political decision-maker who chooses how to act in the face of uncertainty. Moreover, at least in the case of EPA's guidelines, there are indications that the agency will exercise its discretion so as to err on the side of safety. In keeping with this philosophy, for instance, EPA proposes to use the linearized multistage model for high to low dose extrapolations when there is no information to preclude this choice. It is widely recognized that this model provides an upper limit to the risk, but not necessarily a realistic estimate of the actual risk.

OSTP arguably helped to increase the scientific community's control over carcinogenic risk assessment through its efforts to draw up guidelines acceptable to a broad cross-section of scientists. OSTP unquestionably consulted widely with non-governmental scientists in preparing its report, which was incorporated by reference into EPA's guidelines. An unintended side-effect of OSTP's efforts, however, was to heighten the political competition over those aspects of risk assessment that OSTP could not resolve as a matter of scientific consensus. In particular, a controversy developed between EPA and the Office of Management and Budget (OMB) over the content of EPA's risk assessment guidelines. In May 1986 Wendy Lee Gramm, head of OMB's Office of Information and Regulatory Affairs, announced that her staff was 'considering developing more specific guidance for performing risk assessments'.⁶⁰ Lacking any technical expertise in the area, OMB nevertheless seemed prepared to overrule EPA's proposals, although these were developed following extensive public comment and review by the agency's Science Advisory Board. OMB's assertion of control over risk assessment

dramatically negated the view that federal decision-makers are prepared to accept risk assessment as an apolitical, largely scientific exercise.

The chemical industry too has played an active role in the politics of risk assessment. An important test of industry's attitudes to regulatory risk assessment came in 1983 when the manufacturers of urea-formaldehyde foam insulation (UFFI) took the Consumer Product Safety Commission (CPSC) to court over the agency's proposal to ban this product as a potential human carcinogen. A focal point of the lawsuit against CPSC, which was eventually decided in industry's favour on other grounds,⁶¹ was the claim that CPSC's approach to assessing the carcinogenic risk of UFFI was scientifically unsound.

It is worth noting some key features of the strategy adopted by industry to discredit CPSC's decision. UFFI manufacturers were dissatisfied not merely with the quality of CPSC's risk assessment, but with the agency's decision to attempt a quantitative assessment at all on the basis of the available toxicological data. Industry argued, in effect, that formaldehyde should not be treated as a potential carcinogen, let alone be subjected to formal risk assessment. The crux of industry's complaint was that CPSC's decision on UFFI violated scientific norms. By mounting an all-out challenge to CPSC's scientific methods and assumptions, the UFFI manufacturers called into question the agency's competence to resolve issues at the borderline of science and policy. These dynamics suggest that industry's real concern in the area of chemical regulation remains the allocation of power between government agencies and the scientific community, since the latter is seen as more likely to interpret science in ways favourable to industry's interests.

The questions raised by the UFFI manufacturers provided some of the impetus for a further review of the scientific data on formaldehyde at a 'consensus workshop' sponsored by EPA and attended by experts from Europe and the United States.⁶² The UFFI lawsuit thus achieved one of its immediate goals: moving the debate on formaldehyde to a more 'scientific' forum. At issue in the workshop was the correct interpretation of the conflicting data on formaldehyde and the advisability of assessing the compound's carcinogenic potential solely on the basis of the positive animal studies. While no one has questioned the workshop's usefulness as a consensus-building exercise, the proceedings were less effective from the standpoint of providing guidance to EPA. There was no consensus on one of the issues most relevant to the agency's regulatory task, namely, the method that should be used to extrapolate the risk to humans from the available animal evidence. In end effect, then, the workshop was a reminder that, at least in the US policy system,

there are limits to scientific consensus formation as an approach to resolving science policy disputes.

While the chemical industry and its supporters worry that risk assessment leaves too much discretion within the hands of the regulatory agencies, some scientists at least are concerned that the process places undue emphasis on 'cognitive dissonance'⁶³ within science. Weinberg's articulation of these fears is particularly illuminating. He has argued that current attempts by policy-makers to obtain estimates of risk at low exposures transcend the limits of science. By proposing to recognize a new discipline called 'regulatory science', Weinberg asks in effect that scientists be protected against the deconstructive processes of regulatory risk assessment.

I should think that a far more honest and straightforward way of dealing with the intrinsic inability of science to predict the occurrence of rare events is to concede this limitation and not to ask of science or scientists more than they are capable of providing. Instead of asking science for answers to unanswerable questions, regulators should be content with less far-reaching answers... Furthermore, because these same limits apply to litigation, the legal system should recognize, much more explicitly than it has, that science and scientists often have little to say, probably much less than some scientific activists would admit.⁶⁴

The distinction between 'scientists', who presumably understand the limits of science, and 'scientific activists', who refuse to do so, reinforces Weinberg's basic thesis that substantial areas of probabilistic risk assessment do not fall within the domain of science as defined by its professional practitioners.

Peer Review: Rhetoric and Reality

Recognizing the need for scientific legitimacy, the US policy process has cast around for ways of ensuring that science is used in ways that will enhance public confidence. In the early 1970s, complaints by environmentalists about the closed character of expert decision-making led to a tremendous expansion of opportunities for public participation, so that democratic values could be injected into conflicts about the interpretation of science. Federal laws such as the Federal Advisory Committee Act, which mandates more openness in the deliberations of expert bodies, are typical legacies of that period. By the late 1970s, however, a different note entered into the critique of the federal government's analysis of science. Objections stemming from regulated industries

rather than from consumer and environmental activists centred on the 'bad science' allegedly emanating from the agencies.⁶⁵ Critics argued that the science used in regulation was flawed, first, because policy considerations were permitted to taint the evaluation of science, and second, because the agencies lacked sufficient scientific competence. Industry-led efforts to distinguish science from policy (or risk assessment from risk management) were, in part, a response to the first problem; the second has been addressed mainly through demands for more 'peer review' of agency science.

Peer review in the regulatory process derives legitimacy primarily from the fact that this procedure has long been used within science to ensure the validity of new findings and interpretations. Moreover, peer review already has a well-established role in the allocation of research funds by federal granting agencies like the National Science Foundation and the National Institutes of Health. Since regulatory agencies also base many of their decisions on science, it seems reasonable to require that this science too should routinely undergo peer review. The underlying assumption is that the discipline of peer review will help hold governmental science to the same standards as science practised outside government. Timely review, so runs the argument, can help prevent such fiascos as the repudiation of EPA's study of health damage at Love Canal by a blue-ribbon commission appointed to pass on the study's scientific merit.

The structural features that define and motivate peer review in other settings, however, are largely absent in the regulatory context. In the case of both journal articles and grant proposals, for example, there is rarely any question as to what should be reviewed. In regulatory science the reviewable products are much harder to identify. Should agencies submit new test protocols or interpretive guidelines for review, or only completed studies? Should risk assessments be reviewed or only the primary data on which assessments are based? Given the indeterminate character of the 'science' used in agency decisions, the selection of peers, too, becomes a problematic issue. For scientific activity unrelated to policy-making, there is a general agreement that 'peers' are those with relevant disciplinary training and sufficient technical competence to evaluate the matter under review.⁶⁶ In the case of risk assessment, by contrast, it is not even agreed that we are dealing with science in any conventional sense, let alone with science bounded by recognized disciplinary contours and practised by identifiable colleges of scientific peers.

Because of the complicated interplay of science and policy in risk regulation, there is as yet no consensus on how to structure peer review within the policy process. Every agency that makes substantial use of

scientific information has institutionalized some interactions between its own staff and scientists unaffiliated with the agency. These review procedures range from highly formal, legally controlled relationships, such as that between EPA and its Science Advisory Board (SAB), to less structured, *ad hoc* arrangements for examining the scientific basis of regulatory decisions. The potential impact of any kind of regulatory review is, of course, much greater than that of ordinary scientific review. Regulatory decisions carry broader socioeconomic consequences than the research of any individual scientist. Further, rejected articles or grant applications can usually find other sponsors, whereas an unfavourable decision by a regulatory agency can rarely be overturned except through litigation or, exceptionally, through appeals to Congress.

As a highly visible and influential process, regulatory peer review has attracted unfavourable notice from those dissatisfied with policy decisions about risk. Objections focus on the selection of scientific peers. A case in point was the controversy over FDA's use of a so-called Public Board of Inquiry (PBOI) to review the safety of the artificial sweetener aspartame. FDA was criticized for selecting two of the three PBOI members from the Massachusetts Institute of Technology (MIT), thereby permitting an over-representation of a single institutional viewpoint, as well as for putting a nutritionist on the board.⁶⁷ One observer, whose own work included research on aspartame, argued that the scientific disputes about the substance could better have been resolved by trained neuropathologists. But as Vincent Brannigan has noted in his case study of the aspartame PBOI, there was no agreement among the interested parties as to how these scientific issues should be defined.⁶⁸ Hence, it is hardly surprising that there were disagreements about the kind of expertise that should have been represented on the board.

Congress has criticized FDA in the past for wasteful and unnecessary use of advisory committees to resolve issues that were within the competence of the agency's in-house staff.⁶⁹ Such a complaint was recently raised in connection with FDA's regulation of Zomax, an anti-inflammatory drug that not only poses a cancer risk to humans, but has been associated with serious allergic reactions and a number of deaths. In March 1983 Zomax was temporarily removed from the market by its manufacturer, but a subsequent FDA proposal to remarket the drug triggered an investigation by the House Committee on Government Operations.⁷⁰ The committee found fault with several aspects of FDA's decision-making on Zomax, including the referral of the remarketing issue to the agency's Arthritis Advisory Committee. Congressional investigators believed that FDA failed to meet its own

scientific prerequisites for such a referral, since the agency had not received any substantially new information about the drug's chronic effects on health.

In spite of the conflicts and problems associated with regulatory peer review, it is, in the words of one experienced EPA official, a 'growth industry'.⁷¹ Within EPA, for example, demands for SAB's approval of scientific assessments have grown substantially in recent years. In 1981 SAB was reviewing only 15–20 issues a year. The number had risen to 54 for 1985 alone. One reason for the increased popularity of peer review clearly is the desire of political decision-makers to shield themselves against challenge by exhausting all available forms of scientific review. More and more EPA analysts, for example, are turning to SAB because they are aware that SAB's approval could carry weight with both the media and the courts. But regulatory peer review has also won the support of scientists and of private interest groups with quite different stakes in the policy process. Approval by such disparate groups suggests that peer review has a strategic importance extending well beyond concerns for improving the scientific basis of regulatory decisions.

Industry's advocacy of peer review is perhaps easiest to understand in strategic terms. Regulated industries have always insisted that errors in risk management policy arise through the failure of administrative agencies to use 'good science'. A logical next step is to demand that agencies adopt the procedure that scientists themselves use for quality control within their own disciplines. Institutionally, peer review serves industry's interests by transferring some of the discretionary authority to interpret indeterminate science to expert bodies that do not necessarily share the policy biases of federal regulators. The demand for peer review can thus be seen as a natural extension of the same interests that led industry to propose an independent scientific board to carry out risk assessments for all the regulatory agencies.

More puzzling is the readiness of some politicians and public interest activists to embrace the concept of peer review, as they did during the controversy over Todhunter's decision to discount the risks of formaldehyde. At a hearing in the House of Representatives, Todhunter was taken to task by Congressman Gore for not subjecting his analysis of the formaldehyde cancer data to peer review.⁷² Ashford's critique of federal policies on formaldehyde also charged that Todhunter committed a procedural error by not undertaking peer review.⁷³ On the face of it, this apparent endorsement of scientific peer review seems inconsistent with Ashford's special sensitivity to the policy component of risk management decisions.

A closer analysis of the language used by Gore and Ashford, however, indicates that both were using the term 'peer review' differently from industry representatives such as AIHC. As conceived by industry, peer review involves the exposure of scientific analyses done by agency staffs to the scrutiny of scientists situated outside the agencies. Gore and Ashford, by contrast, included within their definition of peer review the internal review of administrative decisions by the agency's own technical staff.⁷⁴

Some of Gore's most pointed questioning of Todhunter focused on the administrator's decision not to consult scientists in the Office of Toxic Substances (OTS) or CAG about his analysis of formaldehyde. Todhunter claimed that there was no need for him to do this because his memorandum represented a policy rather than a scientific decision, and thus was inappropriate for scientific review.⁷⁵ This excuse, however, failed to convince Gore, who insisted that Todhunter's decision was 'science'. In critiquing the administrator's action, Gore used the terms 'peer review' and 'staff review' interchangeably:

However, you speak for the agency on the science question, and if you are going to unilaterally change the science that the agency uses to arrive at a decision, *without peer review, without staff review*, and then you have your judgment called into question by scientists of the eminence of those on the prior panel, then what does that do to the scientific process at EPA?⁷⁶

Ashford, too, collapsed the notions of internal and external review procedures in commenting that 'Todhunter's memorandum was not reviewed by his scientific peers inside or outside the agency'.⁷⁷

Both Ashford and Gore must have recognized that more internal review in the formaldehyde case would very likely have built pressure for overruling Todhunter's assessment. In underplaying the compound's possible carcinogenicity, Todhunter had rejected most of EPA's existing guidelines for assessing chemicals with a carcinogenic potential. Scientists in CAG or OTS, many of whom were involved in developing those guidelines, had a stake in seeing them consistently applied and might well have opposed Todhunter's analysis if called upon to comment on the controversial memorandum. At the very least, their views would have created a record requiring explicit rebuttal, and consultation with EPA's own scientists might even have made Todhunter more sceptical about the analysis advanced by the formaldehyde industry.⁷⁸

Given their professional and political interests, one would expect the scientific community to prefer external peer review over the kind of agency-internal review advocated by Ashford and Gore in the formaldehyde case. Outside review is much more consistent with the image

science seeks to project of itself as an autonomous and politically incorruptible institution. In a congressional hearing following the formaldehyde controversy, Norton Nelson firmly endorsed the expected scientific viewpoint. The immediate object of his remarks was the cancer policy guidelines then under consideration at OSTP. Although these had been informally circulated to independent scientists for review, Nelson did not believe that the final product would achieve scientific credibility:

I have no doubt that there are scientists in the Federal establishment that can do quite as well in terms of the science as scientists outside of the federal establishment, but I do not believe that the credibility required in this important issue can be achieved by cancer policy formulated within a purely administrative framework of OSTP and the federal agencies.⁷⁹

Nelson suggested that for 'credible and technically competent advice' on a revised cancer policy the federal government should turn to an authoritative scientific body, preferably the National Academy of Sciences. Until such advice was forthcoming, Nelson considered it prudent to abide by the cancer policy developed during the Carter administration by the Interagency Regulatory Liaison Group:

The IRLG document, which has been, I believe, disavowed by the present administration, is a perfectly good statement of many of the policies for cancer control. What I am asking is, why should we not stay with the accepted principles and policies which were established until we have undertaken an appropriate review, which I think is very important?

And that review should be done, I sincerely believe, under auspices outside the federal structure.⁸⁰

In using the term 'review', Nelson was referring to the scientific re-evaluation of particular policy-relevant fields (cancer testing, carcinogen risk assessment) rather than to the peer review of any specific document. However, while criticizing OSTP's efforts as inadequate, he held up as a model the guidelines developed by IRLG, another administrative agency within the 'federal structure'. Congressman Ritter of Pennsylvania, a Republican with party loyalties to the Reagan administration, was quick to seize on the apparent contradiction in Nelson's testimony and to shift the debate to the issue of peer review:

Dr Nelson, you mentioned that the current governmental attempt to try to refine or at least expose some possibilities of coming to grips with a policy on cancer, that somehow it was not under the correct auspices; was not proper. I should like to point out that the IRLG, which you seem to hold in very high esteem, was also a governmental body, an interagency body.

It did not bring in peer review. It had a rotating membership. As a matter of fact, the very roughest first document from the OSTP review went out to peer review to people like yourself and members of the environmentalist community.

You know, people have said that the IRLG review occurred behind closed doors and was not open to extensive peer review.⁸¹

Under Ritter's insistent questioning, the issue became not whether OSTP was the right kind of agency to rewrite federal cancer policy, but whether IRLG had conducted a form of peer review that Nelson, as a scientist, would approve. Unable to equate IRLG's practices with the ordinary norms of scientific peer review, Nelson beat a somewhat lame retreat:

Well, I am somewhat familiar with the development of that document . . . It had extensive participation from outside scientists. I think I would agree with you that it should have had outside peer review. It did not.⁸²

The formaldehyde case, then, illustrates the fluidity of the concept of peer review, showing how such a concept can be redefined to suit varying objectives by those with a stake in controlling the discourse of regulation. The Ritter-Nelson exchange indicates how the inherent ambiguity of the word 'review' (review of a scientific field, review of a document) facilitates such manipulation. Of course, this kind of verbal game-playing is easiest with terms that are fairly new in the political discourse. Certainly, the notion of peer review was not widely used in the regulatory context before the 1980s and, as noted earlier, its procedural correlates remain unclear in this setting. Legislative action, such as the adoption of a uniform federal peer review law, could narrow down the uses of the term by rendering certain readings either impossible or implausible. Congress, however, has shown little inclination to propose uniform procedures for regulatory decision-making, and has already mandated several different forms of statutory peer review. Moreover, the preceding discussion suggests that the identification of 'peers' and the structuring of a review process will always be more politically sensitive in the regulatory arena than in the areas of scientific publication or research funding. For these reasons, regulatory 'peer review' may remain a contested concept for the foreseeable future.

Conclusion

The three 'contested boundaries' I have discussed reveal a problem that was largely overlooked in the early attempts to define trans-science

or science policy. The lines between science, policy, and the areas where the two are mixed are difficult to draw not merely because science is indeterminate, but because the effort to make such distinctions is politically charged. How one characterizes an issue on the spectrum between science and policy bears on the way it is ultimately decided, both institutionally and procedurally. If an issue is understood to be scientific, then it can legitimately be resolved by expert panels working with criteria that were never exposed to the full deconstructionist force of the administrative and legal process. Alternatively, if policy elements predominate over scientific ones in a borderline issue, then it is more appropriate to let agency officials and the courts provide the authoritative reading of the disputed technical data, using procedures taken from their respective traditions. The outcome of regulation in a particular case often depends critically on the way decision-making authority is allocated among scientific and political or legal institutions. The classification of issues at the boundary of science and policy, and the procedures used to resolve them, also have a potentially grave impact on the public image of science. Hence scientists, private interest groups and members of the policy establishment all have a stake in the definition of science and non-science, and the vocabulary used by all of these parties remains subject to manipulation or 'essentially contested'.

For scientists, the primary interest in these boundary disputes is to draw the lines between science and policy in ways that best preserve the authority and integrity of science. Scientists have an institutional stake in reducing public interactions between science and the administrative process, since these interactions emphasize the indeterminacy and lack of consensus within science, thereby weakening science's (and the scientist's) claim to cognitive authority. The cases discussed in this article suggest that scientists can adopt two fundamentally different rhetorical strategies to cope with this problem. One approach, perhaps best articulated by Weinberg, is to separate out the areas of maximal uncertainty and conflict and to declare them something other than 'real' science. Weinberg's own proposals for relabelling these boundary areas have included 'trans-science' and, more recently, 'regulatory science'.

Weinberg's approach, though protective of the institutional interests of science, avoids addressing some of the procedural issues of greatest concern to policy-makers. In particular, merely identifying areas where science does not have all the answers fails to settle who should decide questions in these boundary regions, and by what methods or procedures. Scientists actively engaged in the policy process, whether as advisers or as advocates, have not been able to overlook this difficulty. Their

response, exemplified by the positions of Weinstein and Nelson at the cancer policy hearings, has been in some respects almost the opposite of Weinberg's. Both Weinstein and Nelson emphasized, and perhaps exaggerated, the extent of scientific consensus underlying the federal cancer policies developed in the 1970s. Both urged more or less explicitly that revisions in these policies should be undertaken only by accredited scientific organizations using scientific modes of decision-making. In the context of cancer policy, then, Weinstein's and Nelson's remarks were calculated to draw the protective veil of science over many of the disputed issues which others might relegate to the domain of 'regulatory science' or 'science policy'. This approach, too, can be seen as furthering the interests of science, but by enlarging rather than minimizing the role played by science in decisions about risk.

In essence, the position adopted by Nelson and Weinstein with respect to cancer policy is similar to that of scientists in many European regulatory frameworks, where experts wield considerable influence and there is relatively little public debate about the proper limits on their range of inquiry. European officials and the public tend to accept as 'science' any issues that their technical advisory committees are prepared to treat as scientific. There is apparently little concern that policy issues will illegitimately be decided by scientists under the guise of technical decision-making. In the United States, however, the public deconstruction of science has progressed to a point where such trust in expertise is almost unthinkable. Once a policy-relevant area of science has been exposed to deconstruction, as in the case of carcinogen risk assessment, it is virtually impossible to make further decisions in the area without a degree of openness and adversarial debate that is uncommon in European policy systems.

A regulatory process centring on the deconstruction of science threatens the legitimacy of political as well as scientific decision-making. If it is seen that science cannot provide definitive answers to questions about risk, then policy-makers cannot fall back on unassailable technical justifications of their regulatory choices. In the regulatory controversies of the past ten years, especially those involving toxic chemicals, public interest groups have generally argued that in these circumstances regulators should be free to choose among different scientifically plausible interpretations of the evidence, guided primarily by their understanding of their statutory mission. But the strategy of highlighting the indeterminacy of science and the extent of administrative discretion leaves decision-makers vulnerable to charges of using science in arbitrary and capricious ways. The chemical industry has been aggressive in pursuing this line of

attack. Its insistence on 'good science' and its complaints about the technical competence of the federal agencies are consistent with the broader objective of removing as much discretionary power as possible from regulators, who are viewed as captive to environmental interests. Administrators, such as Ruckelshaus, have vacillated between these two positions, hoping, on the one hand, that scientific deliberations will lead to clean, credible resolutions of policy dilemmas, but recognizing, on the other, that such solutions are seldom within the reach of science.

I have suggested here that terms like 'science policy', 'risk assessment' and 'peer review' are used in the regulatory discourse not only for conceptual clarity, but also to advance particular views about the nature of science and its relation to policy. The fluidity of these concepts arises from fundamental social disagreements about the extent to which science and scientists should control decision-making at the frontiers of knowledge. Since these boundary-defining terms affect the allocation of power, their meaning cannot be established independently of the political process. This makes for a certain untidiness in the discourse of risk regulation. Meanings attached to key terms tend to change over time and in the usage of different political groups. New terminology springs up as groups wishing to exert more authority over the interpretation of science find persuasive labels to justify their claims. In a pluralistic society, however, such uses of political rhetoric are entirely familiar, and will trouble only those who expect science to provide rational, apolitical paradigms for the evaluation of risk.

• NOTES

1. See, for example, T. J. Pinch, 'The Sun-Set: The Presentation of Certainty in Scientific Life', *Social Studies of Science*, Vol. 11 (1981), 131-58.

2. D. Nelkin (ed.), *Controversy* (Beverly Hills, CA: Sage Publications, 2nd edn, 1984); R. Brickman, S. Jasanoff and T. Ilgen, *Controlling Chemicals: The Politics of Regulation in Europe and the United States* (Ithaca, NY: Cornell University Press, 1985).

3. M. Mulkay, 'Norms and Ideology in Science', *Social Science Information*, Vol. 15 (1976), 637-56.

4. R. K. Merton, 'The Normative Structure of Science', reprinted as Chapter 13 in *The Sociology of Science* (Chicago: The University of Chicago Press, 1973), 267-78.

5. D. Crane, 'The Gatekeepers of Science: Some Factors Affecting the Selection of Articles for Scientific Journals', *The American Sociologist*, Vol. 32 (1967), 195-201.

6. For a review of the literature pertaining to such structures, see D. E. Chubin, 'Beyond Invisible Colleges: Inspirations and Aspirations of Post-1972 Social Studies of Science', *Sociometrics*, Vol. 7 (1985), 221-54.

7. T. F. Gieryn, G. M. Bevins and S. C. Zehr, 'Professionalization of American Scientists: Public Science in the Creation/Evolution Trials', *American Sociological Review*, Vol. 50 (1985), 392–409; see also Gieryn and A. E. Figert, 'Scientists Protect Their Cognitive Authority: The Status Degradation Ceremony of Sir Cyril Burt', to appear in G. Böhme and N. Stehr (eds), *Sociology of the Sciences Yearbook*, Vol. 10 (1986).

8. In the US legal system, Judge Bazelon of the DC Circuit Court of Appeals has been a noted advocate of open and broadly participatory procedures in decision-making about science. In his view 'the healthiest thing that can happen is to let it all hang out, warts and all'. Statement by D. L. Bazelon, in Office of Technology Assessment, *Impacts of Neuro-Science*, Background Papers (1984), Appendix C, 29.

9. I am indebted to a private communication from B. Wynne for this use of the term 'deconstruct'.

10. See, for example, R. Brickman, 'Science and the Politics of Toxic Chemical Regulation: US and European Contrasts', *Science, Technology and Human Values*, Vol. 9 (1984), 107–11.

11. What is at stake for such scientists is not merely the public characterization of their discipline, but their own status within the scientific profession. To the extent that the public perceives policy-relevant science as not 'real science', its practitioners are in danger of being perceived as not 'real scientists'. This is not a purely theoretical worry, since scientists who participate too actively in policy formulation (for example, as agency employees or professional expert witnesses) often do get read out of the professional scientific community. The boundary work that scientists engage in to determine which policy advisers are true scientists and which are not is a topic that deserves further study, though it lies beyond the scope of this article.

12. For a study of essentially contested terms in the political process, see W. E. Connolly, *The Terms of Political Discourse* (Princeton, NJ: Princeton University Press, 2nd edn, 1983).

13. There are at least two approaches to this kind of line-drawing. One can either try to distinguish science from policy in substantive terms or to differentiate the class of scientists from the class of policy-makers. Following the latter approach, science can be redefined as that which is done by acknowledged scientists. In general, this paper focuses on the former (or substantive) approach to defining the science-policy boundary.

14. A. Weinberg, 'Science and Trans-Science', *Minerva*, Vol. 10 (1972), 209–22.

15. *Ibid.*, 209.

16. *Ibid.*, 215.

17. A. Weinberg, 'Science and Its Limits: The Regulator's Dilemma', *Issues in Science and Technology*, Vol. II, No. 1 (Fall 1985), 67.

18. *Ibid.* (emphasis added).

19. *Ibid.*

20. *Ibid.*, 68. One may wonder whether the practitioners of what Weinberg characterizes as 'regulatory science' would accept this distinction. This inquiry, however, is not pursued further in this paper.

21. *Ibid.*, 220.

22. T. O. McGarity, 'Substantive and Procedural Discretion in Administrative Resolution of Science Policy Questions: Regulating Carcinogens in EPA and OSHA', *The Georgetown Law Journal*, Vol. 67 (1979), 729–810; see also McGarity, 'OSHA's Generic Carcinogen Policy: Rule Making Under Scientific and Legal Uncertainty', in J. D. Nyhart and M. M. Carrow (eds), *Law and Science in Collaboration* (Lexington, MA: Lexington Books, 1983), 55–104.

23. For more detailed accounts of these developments, see S. Jasanoff, 'Science and the Limits of Administrative Rule-Making: Lessons from the OSHA Cancer Policy', *Osgoode Hall Law Journal*, Vol. 20 (1982), 536–61; M. E. Rushefsky, 'Assuming the Conclusions: Risk Assessment in the Development of Cancer Policy', *Politics and the Life Sciences*, Vol. 4 (1985), 31–66. Comparative studies of carcinogen regulation illuminate the degree to which risk assessment in the United States depends on the deconstruction of science: see, for example, S. Jasanoff, *Risk Management and Political Culture* (New York: Russell Sage Foundation, 1986).

24. McGarity, 'Substantive and Procedural Discretion', op. cit. note 22, 732.

25. *Ibid.*, 733.

26. Weinberg, op. cit. note 14, 210–11.

27. McGarity, 'Substantive and Procedural Discretion', op. cit. note 22, 747.

28. Appointed to head EPA early in President Reagan's first term, Anne Gorsuch was forced to resign in 1983, along with many of the agency's top political appointees, following widespread charges of legal violations and administrative misconduct. The episode left EPA badly battered, and morale at the agency was at an all-time low when William Ruckelshaus succeeded Gorsuch as EPA Administrator.

29. See US House of Representatives, Committee on Science and Technology, *Hearing on Formaldehyde: Review of Scientific Basis of EPA's Carcinogenic Risk Assessment* (hereafter cited as *Formaldehyde Hearing*), 97th Congress, 2nd Session (May 1982).

30. N. A. Ashford, C. W. Ryan and C. C. Caldart, 'Law and Science Policy in Federal Regulation of Formaldehyde', *Science*, Vol. 222 (25 November 1983), 894–900.

31. *Ibid.*, 895.

32. The principle that agencies must provide open and rational explanations for departures from prior policies is, of course, well established in US administrative law. See, for example, *Motor Vehicle Manufacturers Association v State Farm Mutual Auto Insurance Co.*, 463 US 29 (1983).

33. Ashford et al., op. cit. note 30, 895.

34. See Brickman et al., op. cit. note 2, 187–217.

35. US House of Representatives, Committee on Energy and Commerce, Subcommittee on Commerce, Transportation and Tourism, *Control of Carcinogens in the Environment* (hereafter cited as *Carcinogens Hearing*), 98th Congress, 1st Session (March 1983), 7.

36. *Ibid.*, 16.

37. *Ibid.*, 70.

38. *Ibid.*, 71.

39. The generic cancer policy was not formally issued until after OSHA promulgated the exposure standard for benzene. However, OSHA relied on the conceptual framework of the cancer policy in developing the benzene standard.

40. *Industrial Union Department, AFL-CIO v American Petroleum Institute*, 448 US 607 (1980).

41. See, for example, AIHC, 'AIHC Proposal for a Science Panel' (mimeo, March 1981); 'Comparative Analysis of HR 6521 (Wampler Bill) and Science Panel Proposal' (March 1980); 'Proposals for Improving the Science Base for Chronic Health Hazard Decision-Making' (Scarsdale, NY: AIHC, December 1981).

42. National Academy of Sciences, *Risk Assessment in the Federal Government: Managing the Process* (Washington, DC: National Academy Press, 1983), 2.

43. L. E. McCray, 'An Anatomy of Risk Assessment: Scientific and Extra-Scientific Components in the Assessment of Scientific Data on Cancer Risks', in National Academy of Sciences, *Risk Assessment in the Federal Government: Managing the Process*, Working Papers (Washington, DC: National Academy Press, 1983), 83–101.

44. 'The risk assessment process requires analytic choices to be made that rest, at least in part, on the policy consideration of whether to be more or less conservative when determining possible public health risks': NAS, *op. cit.* note 42, 139.
45. *Ibid.*, 18.
46. *Ibid.*, 7.
47. *Ibid.*, 49.
48. *Ibid.*, 153.
49. *Ibid.*, 164.
50. W. D. Ruckelshaus, 'Science, Risk, and Public Policy', *Science*, Vol. 221 (9 September 1983), 1027-28.
51. C. Peterson, 'How Much Risk is Too Much?', *Washington Post* (4 February 1985), 8.
52. W. Ruckelshaus, 'Risk in a Free Society', *Risk Analysis*, Vol. 4 (1984), 157-62.
53. Department of Health and Human Services, *Risk Assessment and Risk Management of Toxic Substances* (April 1985).
54. *Ibid.*, 21.
55. *Ibid.*, 29.
56. National Academy of Sciences, *op. cit.* note 42, 166-67.
57. OSTP, 'Chemical Carcinogens: Notice of Review of the Science and Its Associated Principles', *Federal Register*, Vol. 50 (14 March 1985), 10371-422.
58. EPA, 'Proposed Guidelines for Carcinogen Risk Assessment; Request for Comments', *Federal Register*, Vol. 49 (23 November 1984), 46294-301. The guidelines were published in final form almost two years later: see *Federal Register*, Vol. 51 (24 September 1986), 33992-4003.
59. *Federal Register*, Vol. 51 (24 September 1986), 34001.
60. See E. Marshall, 'OMB and Congress at Odds over Cancer Risk Policy', *Science*, Vol. 233 (8 August 1986), 618.
61. *Gulf South Insulation v Consumer Product Safety Commission*, 701 F.2d 1137 (5th Cir. 1983).
62. See National Center for Toxicological Research, Report on the Consensus Workshop on Formaldehyde, Little Rock, Arkansas (3-6 October 1983).
63. Weinberg, *op. cit.* note 17, 67.
64. *Ibid.*, 68.
65. See citations at note 41 above.
66. I. Mitroff and D. Chubin, 'Peer Review at the NSF: A Dialectical Policy Analysis', *Social Studies of Science*, Vol. 9 (1979), 199-232.
67. V. Brannigan, 'The First FDA Public Board of Inquiry: The Aspartame Case', in Nyhart and Carrow, *op. cit.* note 22, 181-202.
68. *Ibid.*, 189.
69. US House of Representatives, Committee on Government Operations, *Report on Use of Advisory Committees by the Food and Drug Administration*, 94th Congress, 2nd Session (January 1976).
70. US House of Representatives, Committee on Government Operations, *Report on FDA's Regulation of Zomax*, 98th Congress, 1st Session (December 1983).
71. Interview with Terry Yosie, EPA, Washington, DC (June 1985).
72. *Formaldehyde Hearing*, 141-43.
73. Ashford et al., *op. cit.* note 30, 897.
74. This usage corresponds to the notion of peer review employed by Congress in the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). The statute authorizes peer

review of EPA studies relating to pesticides by scientists working within or outside the agency. Elsewhere, however, Congress has provided that scientific advisory committees must not include scientists employed by the federal government.

75. Todhunter was here caught in a double bind, though he apparently did not realize it. If his decision on formaldehyde was 'scientific', then it presumably would have benefited from peer review, as urged by Ashford and Core. If however, it was a policy decision, as Todhunter himself claimed, then it clearly went against EPA's established guidelines for carcinogenic risk assessment. As an individual agency official, Todhunter did not have the authority to deviate unilaterally from these guidelines. He should have announced his proposed policy changes openly and submitted them to public comment in accordance with standard US administrative practice. This he failed to do. Thus, whether Todhunter called the decision science or policy, it was procedurally defective.

76. *Formaldehyde Hearing*, 141 (emphasis added); also similar language on 143.

77. Ashford et al., op. cit. note 30, 897.

78. Todhunter was criticized by Norton Nelson of New York University for an overly pro-industry orientation in his analysis of the scientific data on formaldehyde: see *Formaldehyde Hearing*, 29.

79. *Carcinogens Hearing*, 72.

80. Ibid., 73.

81. Ibid., 84.

82. Ibid.

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